

51. Renesas Secure IP (RSIP-E50D)

51.1 Overview

This security engine consists of an isolated subsystem comprising an access management circuit, storage area, encryption/decryption circuit, and random number generation circuit. In combination with the RSIP library, the security engine can prevent eavesdropping (to maintain confidentiality), falsification of information (to ensure integrity), and impersonation (to verify authenticity).

Because key information required for encryption and decryption is stored only in the security engine and all accesses from the outside can be blocked, the security engine enables building a more robust security system.

Table 51.1 lists the security engine specifications. Figure 51.1 shows the security engine block diagram.

Table 51.1 Security engine specifications (1 of 2)

Parameter	Specifications
Access control	Access management circuit In case of irregular access to the security engine due to a tampered program or CPU runaway, this circuit blocks all subsequent accesses and stops data output from the security engine. Interrupt RSIP_TADI is generated.
Symmetric cryptography	AES: Compliant with NIST FIPS PUB 197 - Key length: 128, 192, or 256 bits - Data block size: 128 bits - AES supports the following block cipher mode: - ECB, CBC, CTR: Compliant with NIST SP 800-38A - CCM: Compliant with NIST SP 800-38C - GCM: Compliant with NIST SP 800-38D - XTS: Compliant with IEEE 1619-2007 - AES supports the following authentication algorithms: - CMAC: Compliant with NIST SP 800-38B - GMAC: Compliant with NIST SP 800-38D ChaCha20-Poly1305: Compliant with RFC7539 - Key length: 256 bits - Block size: 512 bits - Support the following algorithms: - ChaCha20 - ChaCha20-Poly1305
Random number generation	128-bit true random number generation circuit
Asymmetric cryptography	RSA - Maximum number of operable bits: 4224 bits - Supported key sizes: 1024 bits, 2048 bits, 3072 bits, and 4096 bits - Signature generation, signature verification, public-key encryption, private-key decryption. ECC - Maximum number of operable bits: 576 bits - Support for curve - NIST P-192, P-224, P-256, P-384, and P-521 - Brain pool P256r1, P384r1, and P512r1 - Ed25519 - secp256k1 - Signature generation, signature verification, key generation.
Message digest computation	HASH - Block size: 512 bits (SHA-224, SHA-256) 1024 bits (SHA-512/224, SHA-512/256, SHA-384, SHA-512) 1152 bits (SHA3-224), 1088 bits (SHA3-256), 832 bits (SHA3-384), 576 bits (SHA3-512), 1344 bits (SHAKE128), 1088 bits (SHAKE256) - Key size: 512 bits or less - HASH supports the following secure hash algorithms: - SHA-224, SHA-256, SHA-512/224, SHA-512/256, SHA-384, and SHA-512: Compliant with FIPS PUB 180-4 - SHA3-224, SHA3-256, SHA3-384, SHA3-512, SHAKE128, and SHAKE256: Compliant with FIPS PUB 202 - HASH supports the following message authentication algorithm: - HMAC: Compliant with FIPS PUB 198

Table 51.1 Security engine specifications (2 of 2)

Parameter	Specifications
Hardware Unique Key	• A read-only, 256-bit Hardware Unique Key (HUK) is exclusively accessible by the security engine access management circuit through a dedicated bus. • Key derivation functions (KDFs) combine the Hardware Unique Key with the key generation information. The derived keys implement the key wrapping for user key secure storage. • The HUK uniqueness prevents the illicit cloning and copying of keys to another MCU of the MCU group. • The HUK itself is stored in wrapped (encrypted, non-plain) format in an isolated memory area. Therefore, it is protected from illicit access and copy.
Application Key Management	• Wrapped keys are only valid within the security engine.
Unique ID	• A read-only, 128-bit ID unique to an MCU (Unique ID) is accessible from the access management circuit. • Key derivation functions (KDFs) combine the Unique ID with the key generation information. Such derived keys are used to unwrap the HUK within the security engine boundary.
OEM boot loader version	• The security engine outputs the current OEM boot loader version (OEM BL Ver.) after successful validation of a firmware update. • MRAM sequencer supports Anti-Rollback Counter Protection using the OEM boot loader version, which can be validated using the security engine.
Decryption-on-the-fly	The security engine outputs the key data of the decryption-on-the-fly IP (DOTF) through a dedicated bus. The key can be set separately for DOTF0 and DOTF1.
Tamper Resistance	Countermeasures available for side-channel attacks, including SPA/DPA and timing attacks.
Module-stop function	Module-stop state can be set to reduce power consumption.
TrustZone Filter	Security and Privilege attribution can be set.

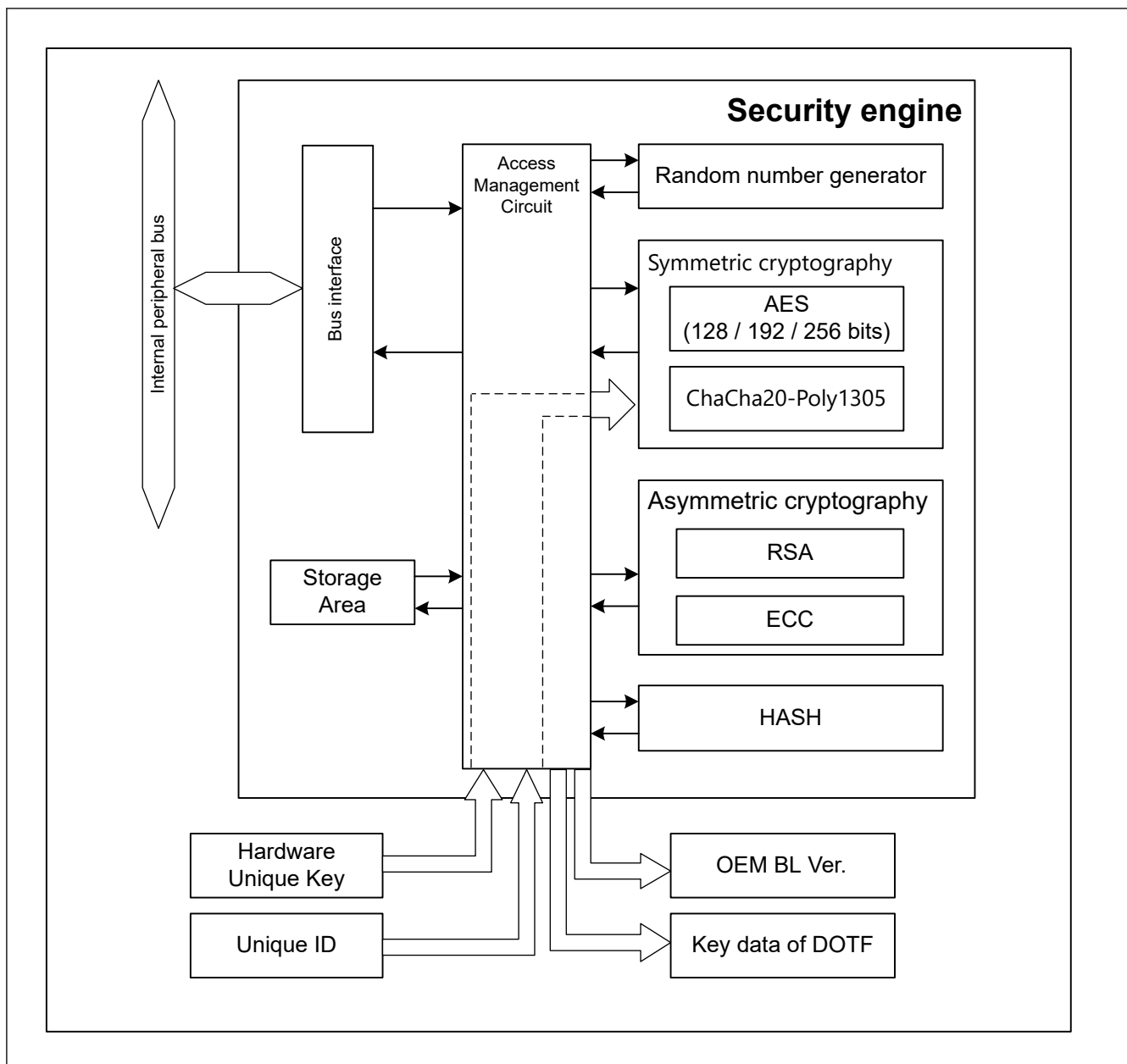


Figure 51.1 Security engine block diagram

51.2 Operation

51.2.1 Symmetric Cryptography

Figure 51.2 shows conceptual block diagram of the symmetric cryptography installed in the security engine.

The symmetric cryptography uses the input key information and converts the plaintext data to ciphertext or ciphertext data to plaintext through the hardware.

The encryption/decryption process can be completed without exposing the key data and the intermediate data of the process outside of the security engine. This process is performed by the encryption/decryption circuit and storage area of the security engine internally.

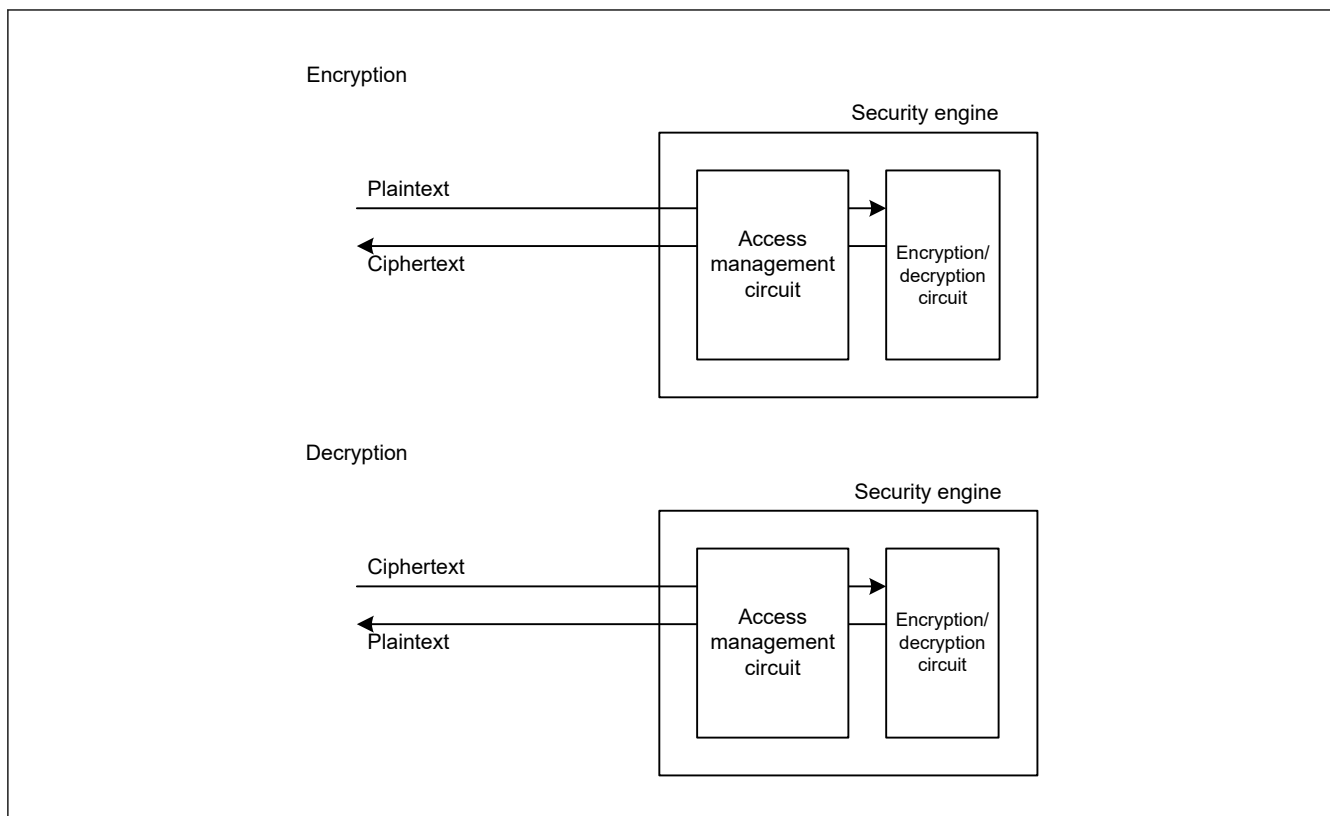


Figure 51.2 Conceptual diagram of the symmetric cryptography

51.2.2 Asymmetric Cryptography

Figure 51.3 shows conceptual block diagram of the asymmetric cryptography installed in the security engine.

The asymmetric cryptography uses the input key information and converts the plaintext data to ciphertext or ciphertext data to plaintext or generates/verifies the signature.

The encryption/decryption or signature generation/verification process can be completed without exposing the key data and the process's intermediate data outside of the security engine. This process is performed by the cryptographic circuit and storage area of the security engine internally.

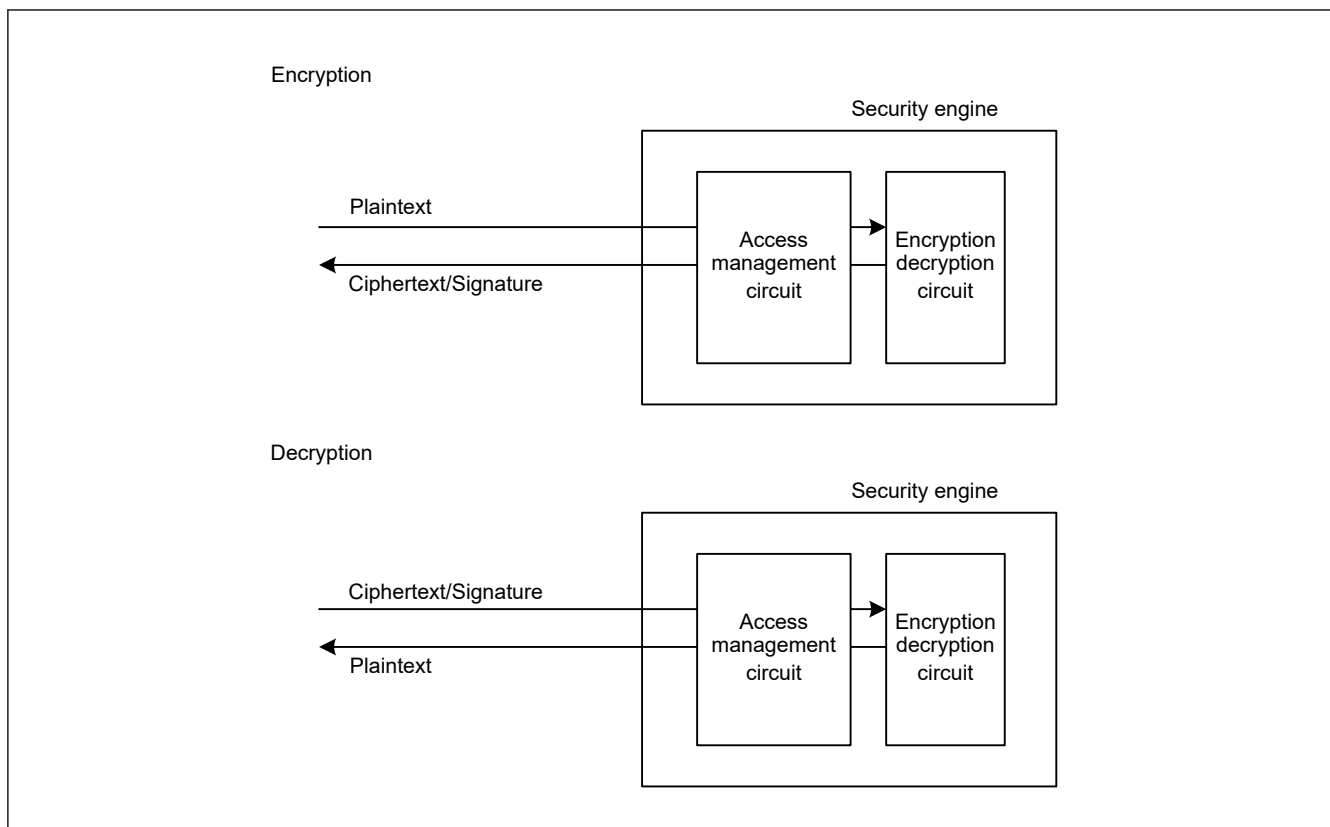


Figure 51.3 Conceptual diagram of the asymmetric cryptography

51.2.3 Hash Generator

Figure 51.4 shows conceptual block diagram of the hash generator installed in the security engine.

The hash generator generates uses the input key information (if required for the hash calculation) and generates the message digest or tag.

The calculation process can be completed without exposing any key data and the intermediate data of the process outside of the security engine. This process is performed by the hash circuit and storage area of the security engine internally.

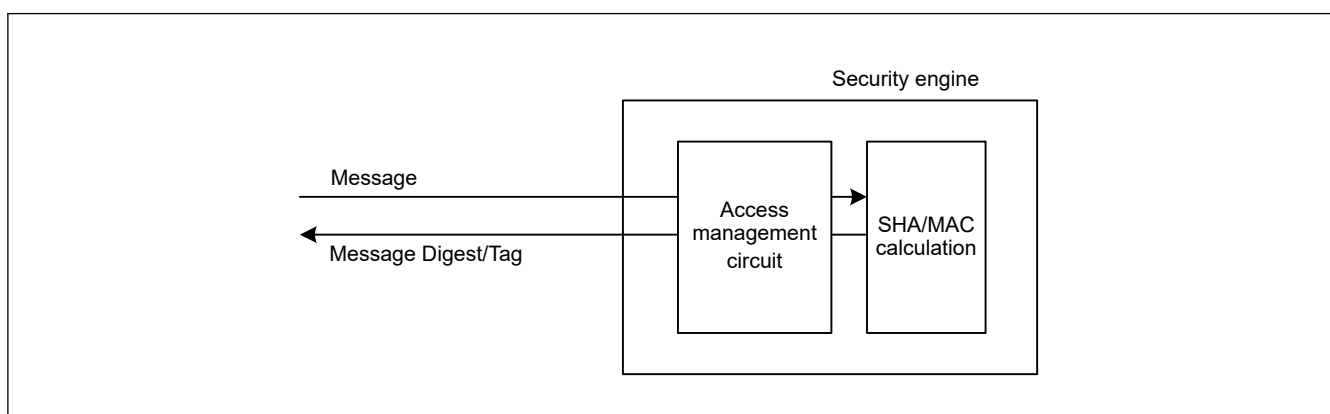


Figure 51.4 Conceptual diagram of the hash generator

51.2.4 Encryption and Decryption

Follow the procedure below to encrypt and decrypt the data:

1. Input the key information to the security engine.
2. Input the target data to the security engine. Plaintext data is converted to ciphertext and ciphertext data to plaintext.
3. Read the converted data.

The encryption/decryption circuit has input and output buffers, and can perform encryption/decryption in parallel with data input/output. Figure 51.5 shows the encryption engine timing.

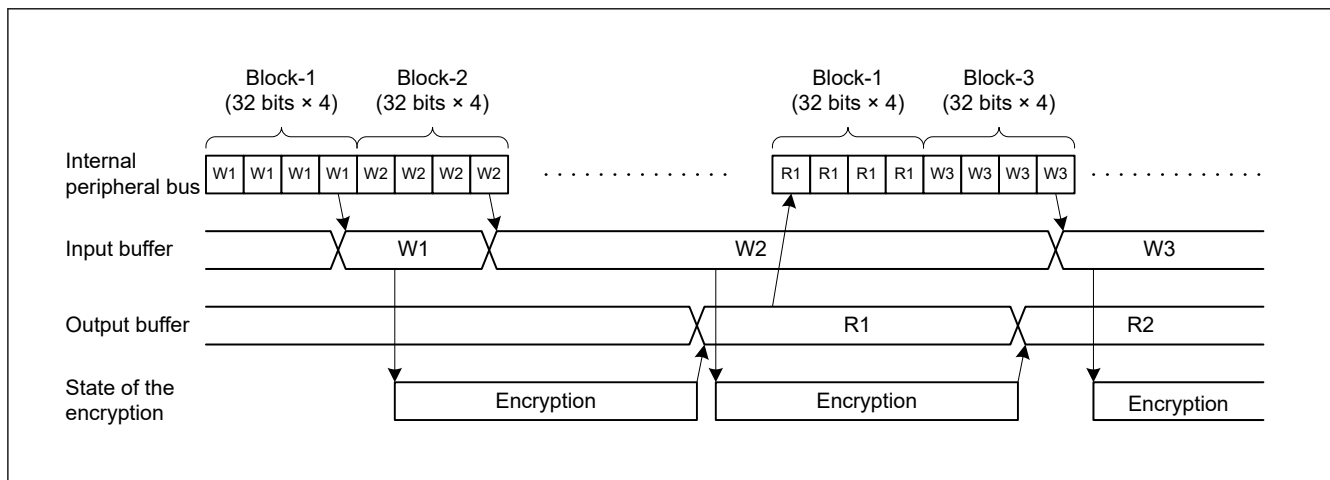


Figure 51.5 Encryption and decryption timing (AES)

51.3 Usage Notes

51.3.1 Software Standby Mode

When Software Standby mode is entered while the security engine is in process, proper processing cannot be resumed after cancelling Software Standby mode. Software Standby mode should therefore be entered while the security engine is not running.

51.3.2 Module-Stop Function Setting

The security engine operation can be disabled or enabled using Module Stop Control Register C (MSTPCRC). The security engine module is initially stopped after reset. Releasing the module-stop state enables access to the registers.

51.3.3 SPA/DPA Protections

SPA/DPA protections can optionally be enabled for encryption/decryption processing.

51.3.4 Restrictions on Open Process

While open process of the API of security engine (R_RSIP_Open) is running, the following restriction applies:

- The CPUCLK0 frequency and the ICLK frequency must be the same.